MODULE 3 · WEEK 1

GitHub & the Cloud (GCP)

Take your project off your laptop: set up Google Cloud Platform, deploy functions, and test them until the check marks turn green.

Lectures 20–23 · GCP Pt. 1–4

This week

1. **Why the cloud** — run code that doesn't depend on your laptop
2. **GCP overview** — the pieces we'll use
3. **Setting up GCP** — project, billing, APIs, permissions
4. **Deploying functions** — your code running in the cloud
5. **Testing** — green check marks mean it works

Why move to the cloud

- Your laptop sleeps, runs out of memory, and isn't always on — the cloud is.
- Cloud functions run on a schedule or on demand, with no machine to babysit.
- This is how data pipelines actually run in industry: code in a repo, executing in the cloud.

Key idea The cloud turns a script on your laptop into a service that runs reliably without you.

GCP overview

- **Project** — the container for everything (resources, billing, permissions).
- **Cloud Functions** — small pieces of code that run on a trigger.
- **APIs & service accounts** — what you enable and authenticate with.
- We connect it all back to your GitHub repo so deployment is reproducible.

Setting up GCP for the project

- Create a project, enable billing, and turn on the APIs the project needs.
- Set up authentication (service account / credentials) — carefully, and never commit keys.
- Follow the setup guide step by step; small config mistakes cause most failures.

```
gcloud auth login
gcloud config set project YOUR_PROJECT_ID
gcloud services enable cloudfunctions.googleapis.com
```

Deploying & testing your functions

- Deploy your function to GCP, then **test it** with sample inputs.
- Watch the logs — they tell you exactly why something failed.
- Iterate until every step passes: that's what the **green check marks** mean.

```
gcloud functions deploy my_func \
  --runtime python311 --trigger-http
```

Key idea Green check marks = your code is live in the cloud and your environment is wired correctly.

Lab this week

- Walk through the GCP setup guide: create the project, enable services, authenticate, and deploy a function.
- Test your deployed functions and confirm everything reports success.

In the labs repo: `Module3/Week1_GitHub_Cloud/GCP_SETUP.md` · `Module3/Week1_GitHub_Cloud/GCP_Setup_Guide_Students.ipynb`

Key takeaways

- The cloud runs your code reliably without depending on your laptop.
- A GCP project ties together resources, billing, APIs, and permissions.
- Deploy functions, test with real inputs, and read the logs when they fail.
- Never commit credentials — authenticate with service accounts and env vars.
- Green check marks confirm the whole pipeline is wired up.